

Application Note

Bulk capacitor sizing for DC motor drive applications



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ABSTRACT

Appropriate local bulk capacitance is an important factor in motor drive system design. Having more bulk capacitance is generally beneficial, while the disadvantages are increased cost and physical size. This application note discusses general guidelines for selecting the amount of capacitance needed in a motor drive system.

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1 Introduction

Direct current (DC) motors are found in a wide variety of applications in automotive, industrial, and consumer products. When switched on or off, including during pulse-width modulation (PWM) operation, the motor current can change significantly. These current changes can create issues such as supply voltage variations and electromagnetic interference for nearby electronics.

It is common to include large "bulk" capacitors as part of the motor driver design. These bulk capacitors act as a local reservoir of electrical charge to smooth out the motor current variation.

Designers look for guidance on the appropriate values of bulk capacitance. In the following discussion, we will look for methods to select a proper value of capacitance based on knowledge of the motor and driver parameters. The nomenclature is based on brushed DC motors, but the same principals apply to other types, including brushless DC (BLDC) and stepper motors.

Figure showing bulk capacitors on one of the EVMs.



Figure 1-1. Typical motor driver board showing large bulk capacitor

2 Background and theory

2.1 Factors affecting bulk capacitor sizing

The amount of bulk capacitance needed depends on a variety of factors including:

- The highest current required by the motor system
- The power supply's type, capacitance, and ability to source current
- The amount of inductance between the power supply and motor system
- Method of operation, whether continuous or using pulse width modulation
- The acceptable supply voltage ripple
- Type of motor (brushed DC, brushless DC, stepper) and motor characteristics
- The motor startup and braking methods

In an ideal DC motor system, there would be no impedance between the motor drive circuit and the power source, which could be modeled as an ideal constant voltage source. In this ideal case, there would not be any variation in the motor supply voltage. However, in a real practical system, the inductance between the power supply and motor drive system will limit the rate at which current from the power supply can change. There may also be filtering inductance on the driver board or in the power distribution network. This inductance between the power supply, often a 12V battery in automotive systems, and the motor drive voltage V_M , is modeled as a lumped inductance on the positive and ground connections labeled “Parasitic Wire Inductance” in Figure 1.

If the local bulk capacitance is too small, the system will respond to excessive current demands or dumps from the motor with a change in motor supply voltage. When adequate bulk capacitance is used, the motor supply voltage remains stable and high current variations can be quickly supplied.

Figure 10-1 in SLVSEA2c

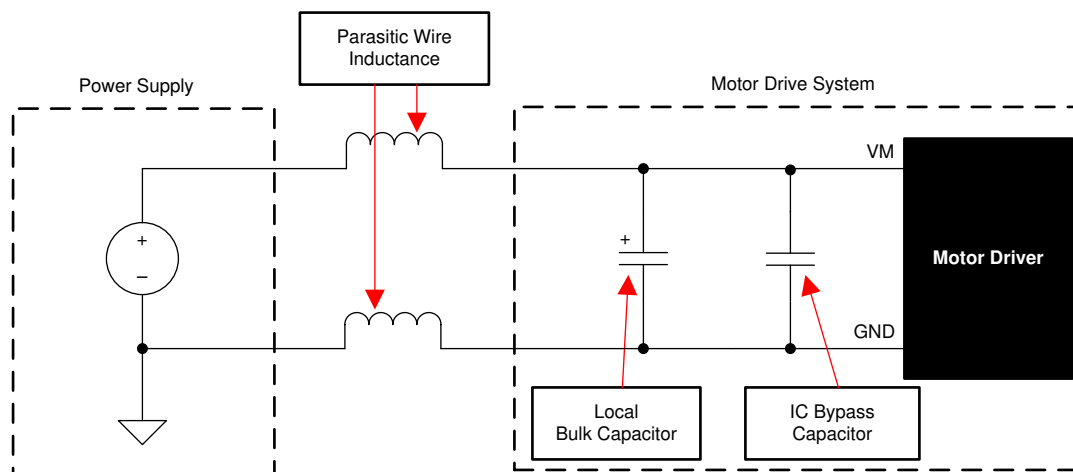


Figure 2-1. Motor drive model, from DRV8718-Q1 datasheet

The motor driver data sheet may provide a recommended *minimum* value, but system level testing is required to determine the appropriately sized bulk capacitor. Figure 2 shows an example of the recommendations in a TI motor driver datasheet.

2.2 Pulse width modulation

If pulse-width modulation (PWM) is used to control the effective motor voltage, the bulk capacitance on the motor supply becomes even more important. Otherwise the current variation during PWM operation can cause voltage ripple which can propagate throughout the system, causing electromagnetic interference with other components. Figure 3 shows typical waveforms in a motor drive system with PWM at 20 kHz. The red trace marked “MOTOR

CURRENT" shows the effect of rising current as the motor is activated, with ripple current corresponding to the PWM frequency.

Figure 9-3 in SLVSEA2c

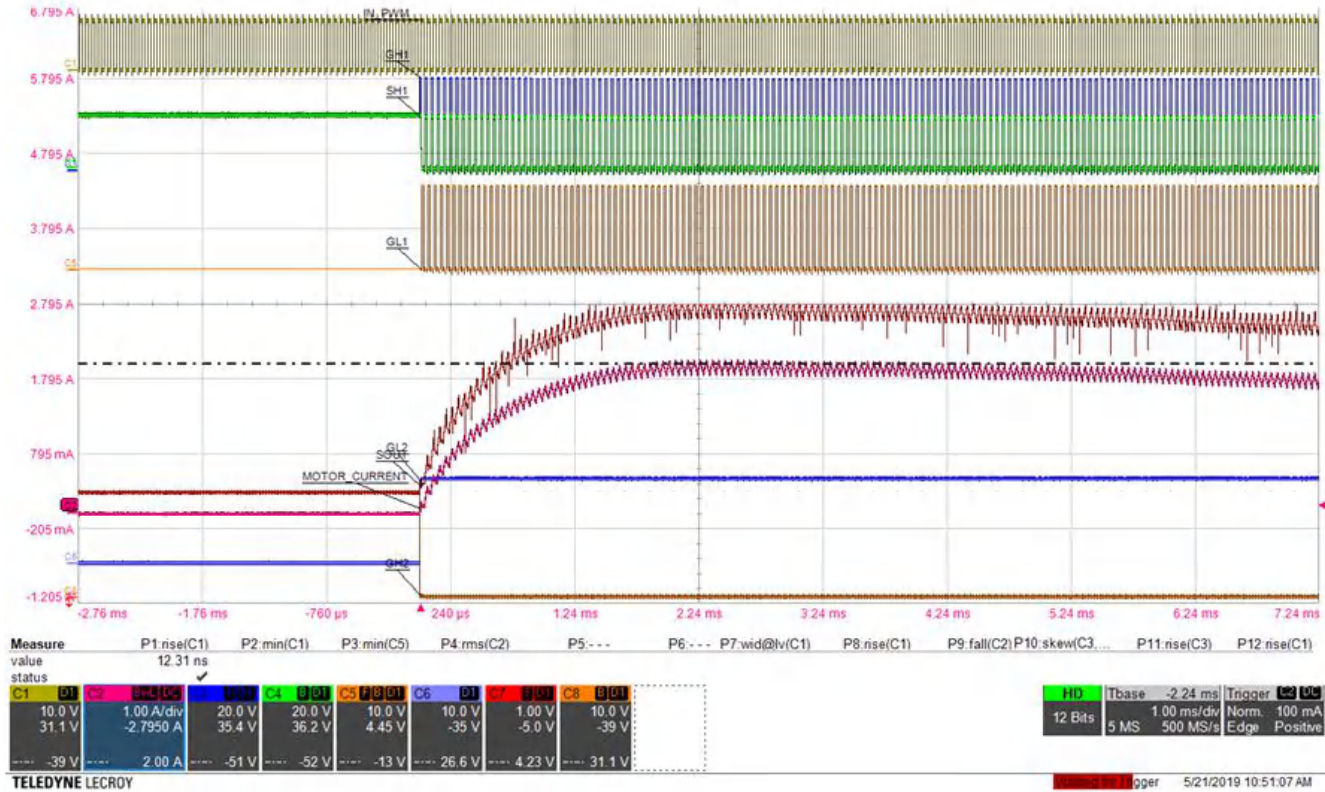


Figure 2-2. Typical waveforms in a PWM motor drive, from DRV8718-Q1 datasheet

In this example, the scale is 1A per major division of the current trace, so the variation in motor current is on the order of 200 mA due to the PWM switching. This is approximately 10% of the peak motor current of about 2 amps.

2.3 Estimating motor current variation

During PWM operation, the motor current varies around an average value, with the current increasing during the 'on' portion of the PWM period, and decreasing during the 'off' portion. Modeling the motor winding as an inductance L_m in series with a resistance R_m , the increasing current is:

$$i_{inc}(t) = i_{min} + [i_f - i_{min}] \times (1 - e^{-t/\tau}) \quad (1)$$

Where i_f is the final steady-state current at 100% duty cycle, typically $PVDD/R_m$ and i_{min} is the current at the beginning of the PWM 'on' portion, and tau is the motor electrical time constant, L_m/R_m .

Similarly for the decreasing current:

$$i_{dec}(t) = i_{max} + [0 - i_{max}] \times (1 - e^{-t/\tau}) = i_{max} \times e^{-t/\tau} \quad (2)$$

Where i_{max} is the current at the beginning of the PWM 'off' portion.

The motor current variation $\Delta I_{motor} = i_{max} - i_{min}$ is the motor current variation over each PWM cycle.

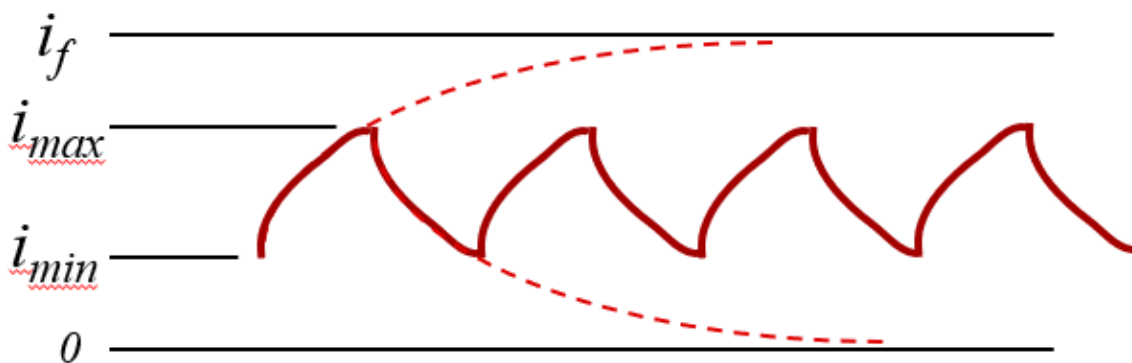


Figure 2-3. Motor current variation during PWM operation , ideal capacitors

2.4 "Rule-of-thumb" calculations assuming ideal capacitors

As discussed above, a large value of bulk capacitance is desired to provide a constant motor supply voltage during current transitions, such as motor start-up, changes in load torque, or PWM operation. But we would like to have a working estimate of the needed capacitance so that we do not over-design the bulk capacitance, leading to high system cost and excessive board size. We can use a rule-of-thumb method to find an appropriate capacitor size based on the expected load current variation and allowable motor supply voltage variation.

An initial estimate of the appropriate bulk capacitance based on *ideal* capacitors is:

$$C_{BULK} > \Delta I_{MOTOR} \times T_{PWM} / \Delta V_{SUPPLY} \quad (3)$$

Where C_{BULK} is the bulk capacitance, ΔI_{MOTOR} is the expected variation in motor current, and ΔV_{SUPPLY} is the allowable variation in the motor supply voltage..

This inequality is based on assumptions:

Ideal capacitors with zero equivalent series resistance (ESR)

A motor time constant L_m/R_m significantly longer than the PWM period

Negligible change in supply current through the parasitic wire inductance

The supply voltage variation induced by the varying motor current during PWM can be expected to be a maximum when the PWM duty cycle is 50%.

So as an example, assume we have motor current variation during PWM of about $\Delta I_{MOTOR} = 200$ mA; this is approximately what is shown in Figure 3; a simplified sketch is shown in Figure 5. For a typical PWM frequency of 20 kHz, so T_{PWM} is 50 microseconds.

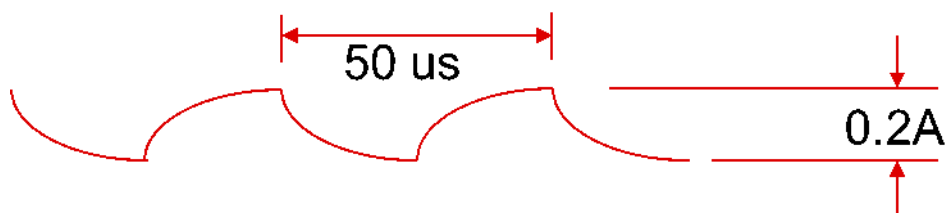


Figure 2-4. Simplified example motor current waveform during PWM

If we want to keep the motor supply voltage variation not more than $\Delta V_{SUPPLY} = 100$ mV, we can estimate the required minimum bulk capacitance as:

$$C_{BULK} \geq 200\text{mA} \times 50\mu\text{s} / 0.1\text{V} = 10\mu\text{A} \cdot \text{s} / 0.1\text{V} = 100\mu\text{F} \quad (4)$$

This is an estimate, not a detailed analysis. We are neglecting several factors such as the effective series resistance (ESR) of the bulk capacitance, the non-infinite impedance of the inductance to the battery, the non-linear current variation shape, etc. However, it gives a reasonable bulk capacitance value as a starting point for more rigorous system analysis.

The results for the ideal simulations are compared to what the ideal inequality predicts in Figure 6. The simulated voltage ripple is somewhat smaller than the ideal estimate predicts, but overall the predictions are approximately correct, especially for higher values of bulk capacitance.

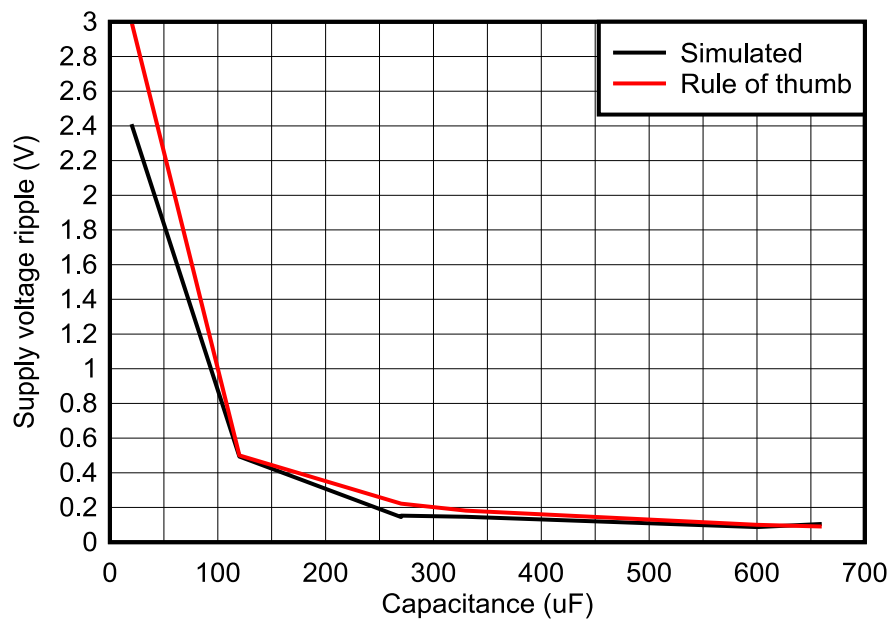


Figure 2-5. Comparison of ideal capacitor (ESR=0) rule of thumb and simulated results

However, as we see in the following discussion, real-life measurements indicate the ideal results are off by as much as a factor of 3, which means the non-ideal capacitance ESR should not be neglected.

3 Real-world (non-ideal) results

Laboratory measurements show that the calculations and rule-of-thumb discussed above do not accurately match real-world results, due to the non-ideal characteristics of typical bulk capacitors.

3.1 Example measurements

Example measurements

Using a DRV8718-Q1 EVM, we can measure examples of the effect of bulk capacitance. We used a 50% duty cycle PWM signal with a 12V supply, and a load of 3 Ohms in series with a 470 uH inductor to simulate a typical DC motor circuit. The L/R time constant is 157 microseconds, about three times the PWM frequency.

Figure 7 shows the case where the bulk capacitance is 270 uF. Using the estimate for ideal capacitors, we would expect about 80 mV of variation in PVDD due to the capacitor charging and discharging during PWM. Neglecting the quick shift during each transition, the voltage ramp does have an amplitude of about 80 mV, as indicated by the dashed cursors.

However, the total variation in PVDD in Figure 7 is about 160 mV, significantly more than predicted by the ideal capacitor estimate. There is a sharp shift in the voltage at the beginning of each transition, which is due to the non-ideal properties of the bulk capacitors. The equivalent series resistance (ESR) of electrolytic capacitors can account for this quick voltage shift.

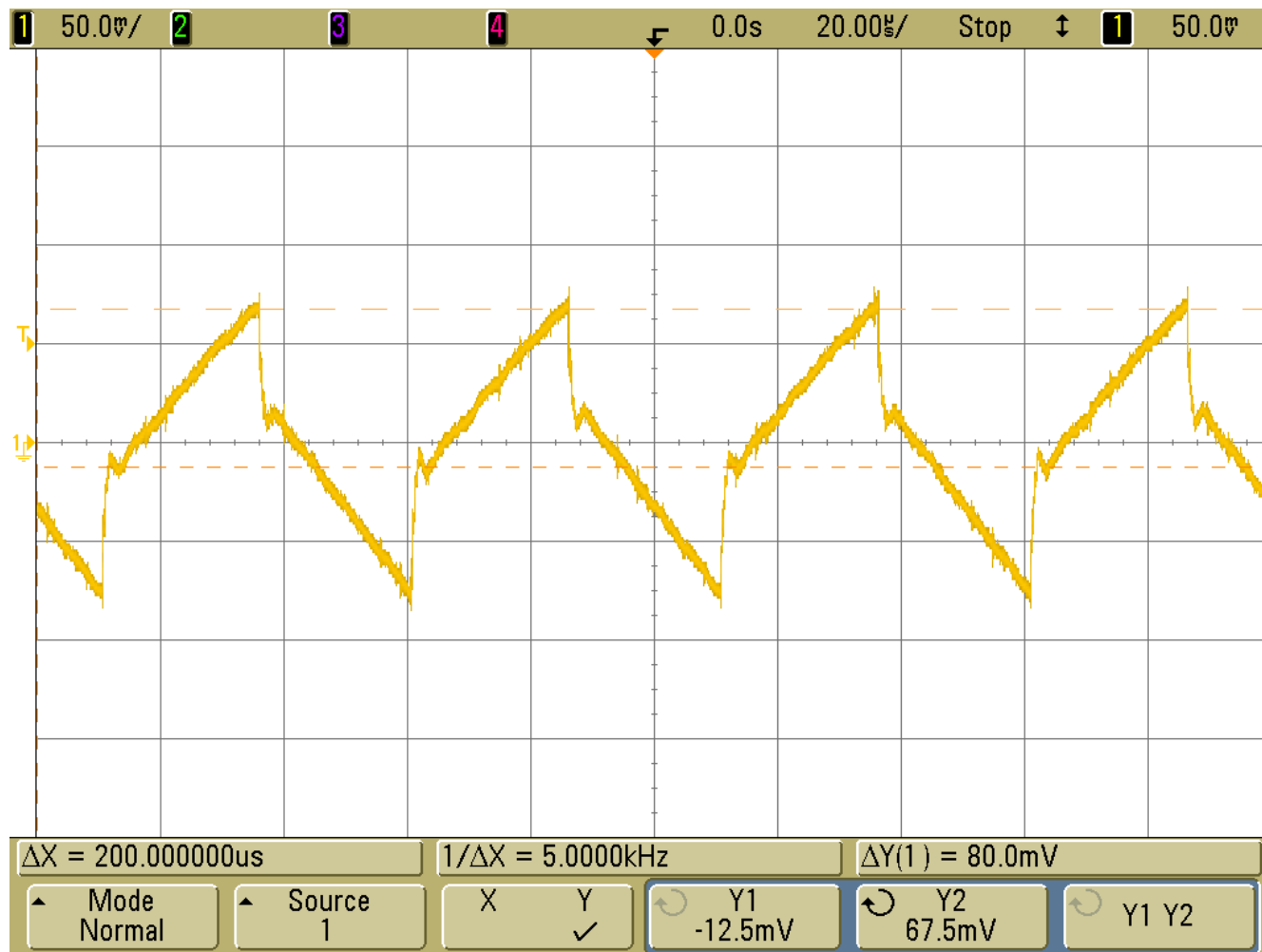


Figure 3-1. PVDD variation with 270 uF bulk capacitance, measured results during PWM

The variation on PVDD consists of mainly two components, the voltage due to the charging and discharging of the bulk capacitance, and the voltage across the non-ideal equivalent series resistance (ESR) of the capacitors.

The equivalent series resistance (ESR) of real capacitors is one measure of how non-ideal their characteristics are. Low ESR capacitors are typically more expensive, but may provide benefits in terms of reduced voltage ripple. This is a common topic in power supply design, for example, see the TI application note “Output Ripple Voltage for Buck Switching Regulator”, <https://www.ti.com/lit/SLVA630>.

With simulation, we can see what the ripple looks like with an ideal capacitor having zero ESR, and also model the real-world capacitor by adding a resistance in series to represent the ESR.

First let's look at how the current flows in the capacitor, and thus how the voltage varies across the ESR we model as part of the capacitor. During the 'on' part of the PWM cycle, the capacitor acts as a source of motor current, so the voltage drop across the ESR subtracts from the internal “ideal” voltage stored in the capacitor. Thus the voltage across the capacitor, including the ESR, is lower than the internally stored voltage. During the 'off' part of the PWM cycle, the capacitor acts as a current sink, charging up the internal voltage. During this charging, the voltage across the capacitor, including the ESR is higher than the internally stored voltage. The motor and capacitor currents are shown in Figure 8.

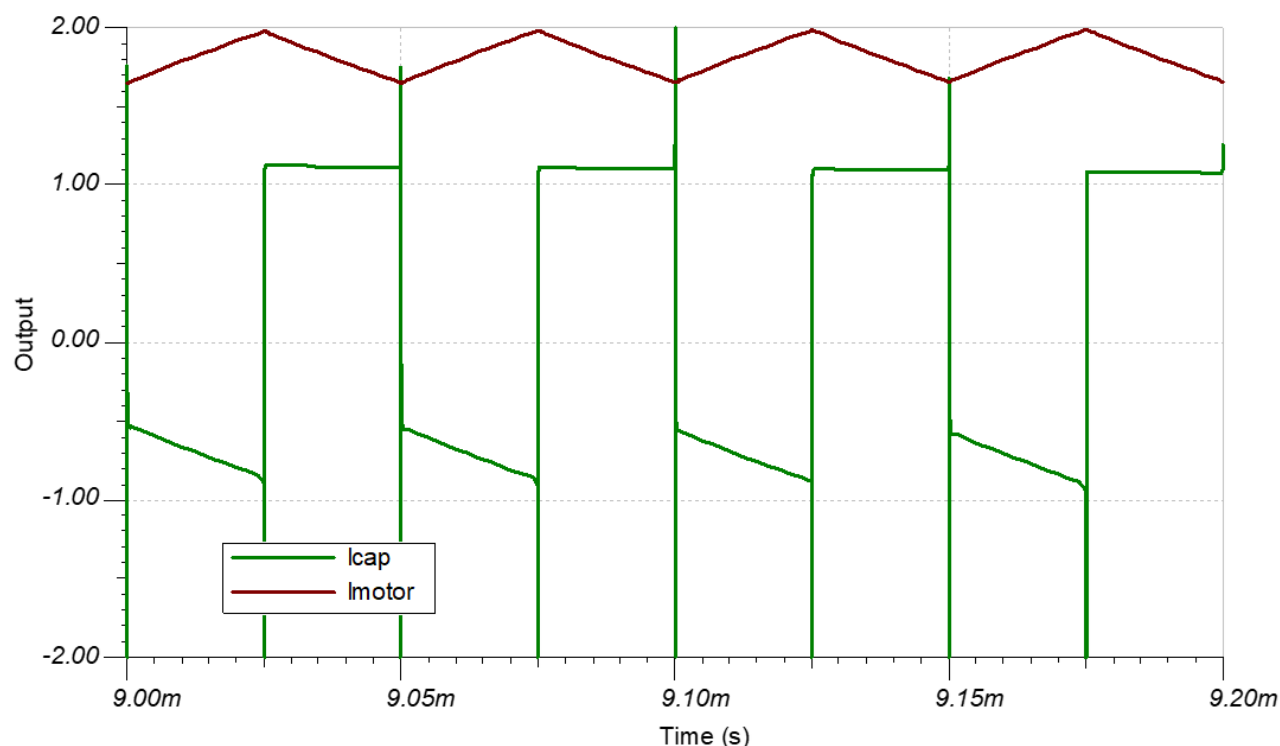


Figure 3-2. Motor current (red) and capacitor current (green) during PWM

In Figure 9 a TINA simulation shows the voltage variation due to PWM of the MOSFET T1, with a bulk capacitance of 270 μ F and zero ESR, an ideal case. The voltage variation on PVDD is about 100 mV. When

50 mΩ of ESR is modeled in Figure 10, the voltage variation increases to about 200 mV, and the shape of the voltage ripple looks more like the real-world measured case.

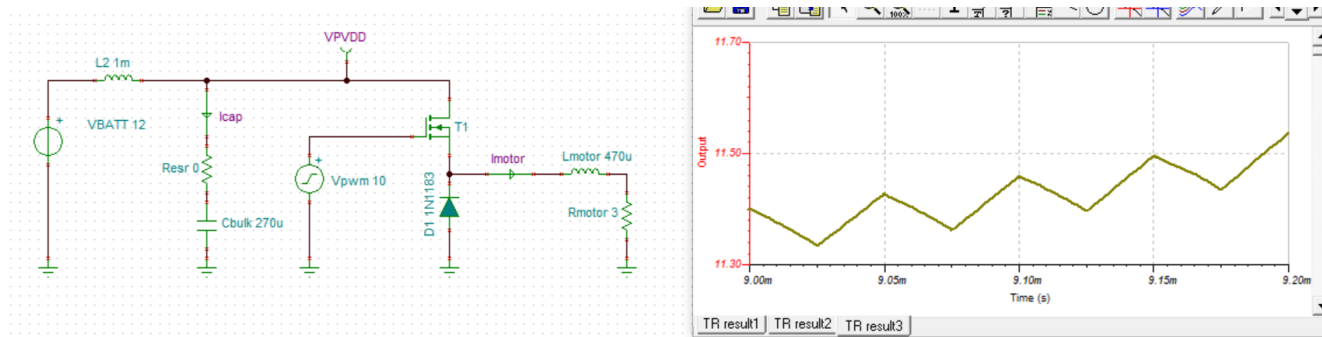


Figure 3-3. TINA simulation with zero ESR. This simulation largely agrees with the ideal capacitance equations.

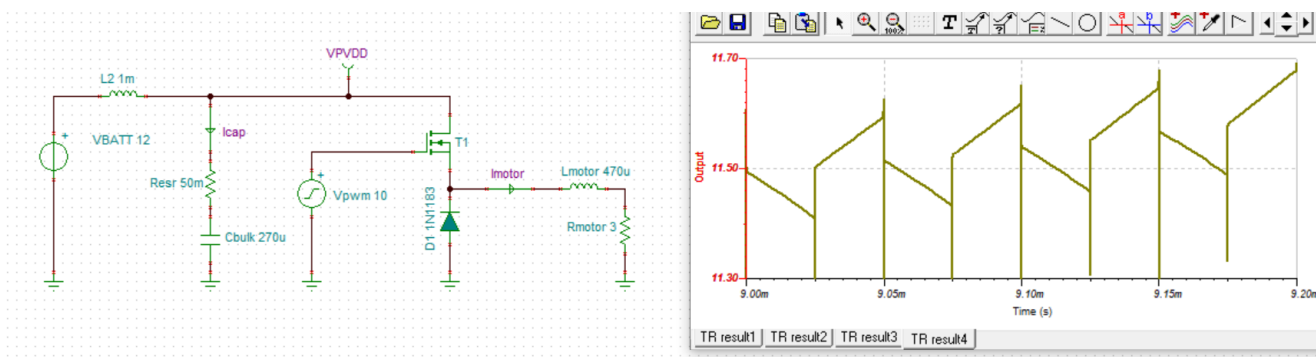


Figure 3-4. TINA simulation with 50mΩ ESR. Note how adding ESR makes the waveform look more like the actual results in Figure 7.

Another case is shown in Figure 11 where the bulk capacitance has been increased to 600 uF by adding a 330 uF capacitor in parallel with the 270 uF of the previous case. The ideal capacitor estimate given above would predict a voltage variation of about 36 mV. The measured value is about 57 mV; so while the voltage variation

has been reduced by adding more bulk capacitance, it has not been reduced as much as predicted by the ideal estimate.

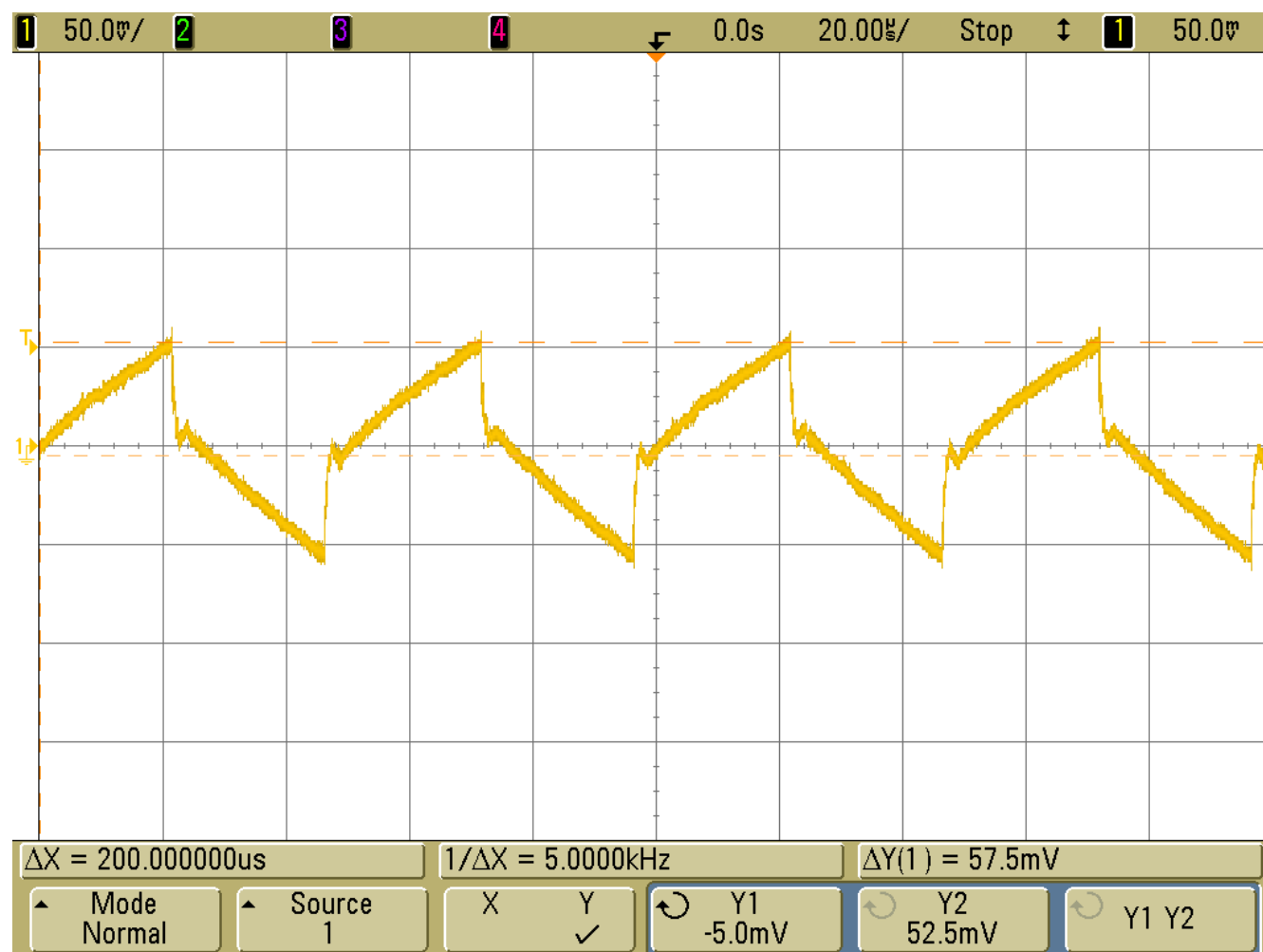


Figure 3-5. PVDD variation with 600 uF bulk capacitance

Figure 12 shows one more example, with the bulk capacitance reduced to 120 uF. Now the voltage ripple increases to almost 500 mV, and the shape is dominated by the charging and discharging of the capacitance, and the effect of the ESR is less evident.

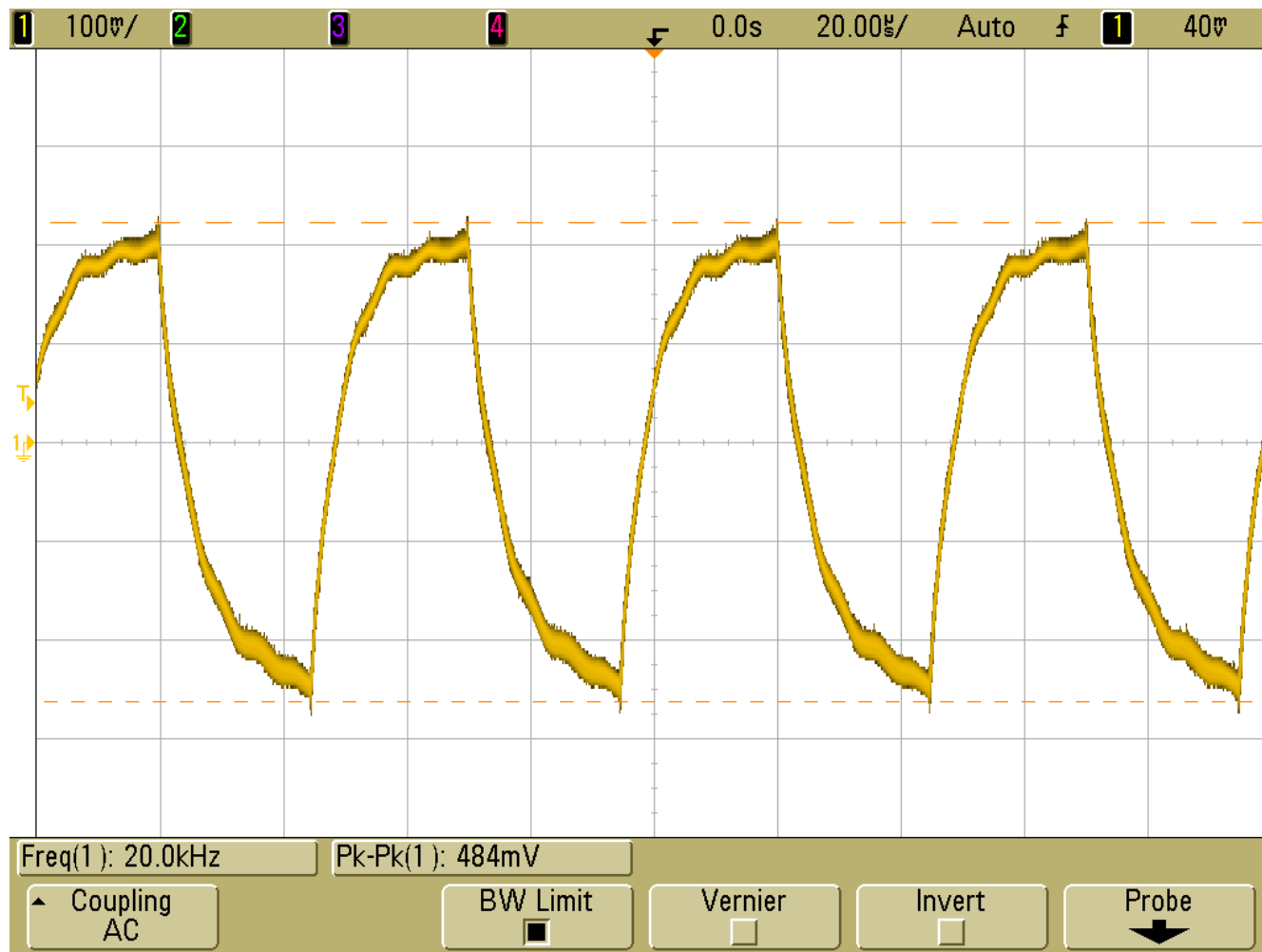


Figure 3-6. PVDD variation with 120 uF bulk capacitance

3.2 Revised practical rule-of-thumb

Based on these real-world observations, we modify the ideal rule-of-thumb given previously to account for the non-ideal characteristics of the capacitor:

$$C_{BULK} > k \cdot \Delta I_{MOTOR} \cdot T_{PWM} / \Delta V_{SUPPLY} \quad (5)$$

Where:

C_{BULK} is the bulk capacitance

k is a scale factor based on the ESR for typical capacitors in this type of application; based on lab measurements above with DRV8718 EVM, k of about 3 is practical.

ΔI_{MOTOR} is the expected variation in motor current, $i_{max} - i_{min}$

T_{PWM} is the PWM period which is the reciprocal of the PWM frequency

ΔV_{SUPPLY} is the allowable variation in the motor supply voltage

Figure 13 plots several data points and applies this rule-of-thumb, showing relatively good agreement.

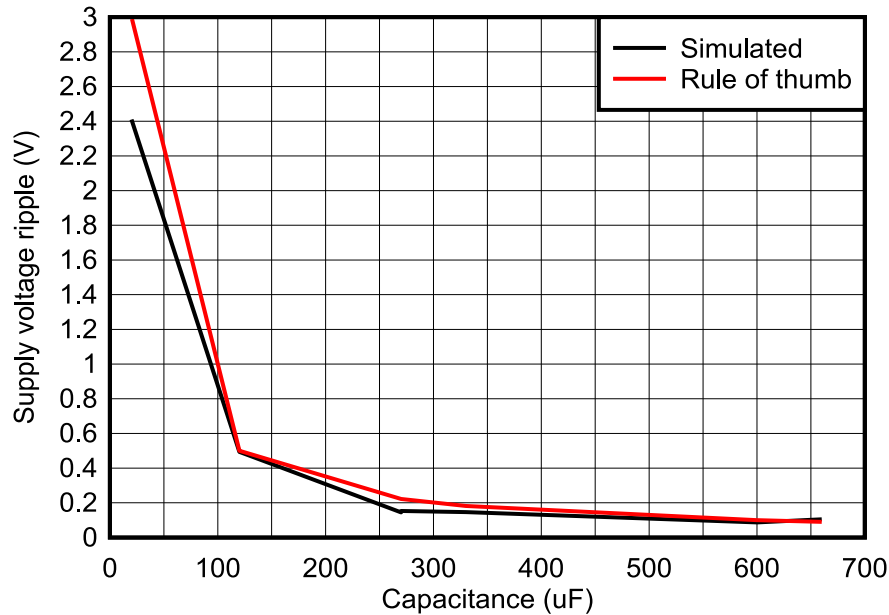


Figure 3-7. Measured results and "3x" rule of thumb, accounting for real-world non-zero ESR values of electrolytic capacitors

3.3 Other considerations

The discussion has been simplified to allow rapid estimates of bulk capacitance. Several other factors affect the system performance and should be taken into account when selecting bulk capacitors for a motor drive design. These include:

Frequency response – capacitor frequency response varies between types of capacitors and between capacitors with different values. It is often recommended to provide more than one value of capacitor on the motor power supply. The larger value capacitors typically provide the best low-frequency response, while smaller value capacitors provide better high-frequency response.

4 Summary

In a real DC motor drive system, bulk capacitors are a common necessity. Although final system performance requires detailed analysis and practical testing, we can use rules-of-thumb and simple simulations to estimate the bulk capacitor sizing as a starting point.

5 References

- “Input and Output Capacitor Selection”, TI Application Note, <https://www.ti.com/lit/pdf/slta055>
- “Output Ripple Voltage for Buck Switching Regulator”, TI Application Note, <https://www.ti.com/lit/SLVA630>
- “Voltage margin and bulk capacitance”, TI training video: <https://training.ti.com/ti-precision-labs-motor-drivers-voltage-margin-and-bulk-capacitance>
- “Real world brushed DC motor bulk capacitance calculations”, Electronics.Stack Exchange website, <https://electronics.stackexchange.com/questions/483835/real-world-brushed-dc-motor-bulk-capacitance-calculations>

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